

# Bihan Banerjee

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## EXPERIENCE

### Full Stack Intern

May 2026 – Present

*Reliance Industries Ltd*

*Jamnagar, India*

- Developing full-stack internal tooling for **Jamnagar refinery complex**, the **world's largest oil refinery**, using **Angular**, **Node.js**, and **MS SQL Server** for operations and workflow data management.

### MLOps Intern

June 2025 – July 2025

*DCG AILABS Academy Pvt Ltd*

*Kolkata, India*

- Built and containerized a **real-time sentiment analysis system** on **50K+ IMDb reviews** using Flask, TF-IDF, Docker, and **Kafka/Zookeeper** streaming pipelines, achieving **89% accuracy**.

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, C/C++, SQL

**Cybersecurity:** Burp Suite, Kali Linux, Wireshark, Sengrep, Bandit, Trivy, OWASP ZAP, SonarQube, Metasploit

**Frameworks & Infra:** React, Next.js, Node.js, Flask, Express, Tailwind CSS, Docker, Kubernetes, Helm

**Databases & Cloud:** MongoDB, Kafka, AWS (EKS, ECR, S3), Terraform, Git

## PROJECTS

### GuardOps – Production-Grade DevSecOps CLI | *Python, Docker, Kubernetes, Terraform, AWS*

2026 – Present

- Engineered a **CLI-driven secure delivery pipeline** integrating **Sengrep, Bandit, Trivy, SonarQube, and OWASP ZAP** across pre-deploy SAST and post-deploy DAST stages, blocking pipelines on HIGH+ findings.
- Deployed to **AWS EKS via Helm** with Terraform-managed VPC/ECR/IAM infrastructure, GitHub OIDC (no static keys), automatic rollback on CRITICAL DAST findings, and **167+ tests** across 5 test suites.
- Built a full **Prometheus + Grafana observability stack** with custom application metrics, ServiceMonitor scraping, and PromQL dashboards; published to **PyPI** with **1,700+ downloads** across 8 versioned releases.

### CyberX – AI-Powered Cybersecurity Simulator | *Python, MERN, Reinforcement Learning*

2025 – 2026

- Architected a modular **red-team/blue-team cybersecurity framework** with a custom pentesting suite (**25+ tools**) spanning cryptographic, forensic, and defensive monitoring modules.
- Built **real-time attack intelligence dashboards** with geospatial globe-based visualization, handling **1,000+ live events** at **<2s refresh latency** and system metrics at **1s sampling (<5% overhead)**.
- Developed **PPO-based attacker-defender agents** via adversarial self-play across **500K+ timesteps**, simulating **100-step episodes** using honeypot telemetry to improve detection rates.

### FortiScan – Vulnerability Scanner | *React, Flask, TypeScript, MongoDB*

2025

- Designed an AI-driven platform to detect and **auto-remediate vulnerabilities** in LLM-generated Python/JS code using **CodeT5, GPT-Neo, and CodeGPT-small**.
- Integrated **Bandit, Sengrep, SonarQube** for multi-layered analysis mapped to **100+ CWE vulnerability patterns**, enabling comprehensive detection coverage.
- Built a scalable **React (TypeScript) + Flask system** with multi-file uploads, real-time dashboards, and AI-based patch generation with **diff visualization**, reducing insecure patterns by **70–85%**.

## PATENTS

### Intelligent Real-Time Security Analysis Platform for AI-Generated Code

2026

*Intellectual Property India (Application Number: 202641027876)*

*Patent*

- Filed **IDE-integrated** system for real-time **AI-generated code analysis** using **multi-model vulnerability fingerprinting** to capture LLM-specific security patterns.
- Built an AI remediation engine with **context-aware secure code fixes**, adaptive alert optimization, and **temporal evolution tracking** via probabilistic methods.

## EDUCATION

### Vellore Institute of Technology (VIT)

Vellore, India

*B.Tech CSE (Information Security) | CGPA: 8.34*

*2023 – Present (Expected 2027)*

### Garden High School (CISCE)

Kolkata, India

*Class XII: 91.25% | Class X: 98.6%*

*2008-2023*

## EXTRACURRICULAR ACTIVITIES

### General Secretary, Mozilla Firefox Club VIT

July 2024 – Present

*Vellore Institute of Technology*

*Vellore, India*

- Led planning and execution of technical events and hackathons for **300+ students**, coordinating sponsorships and logistics.
- Anchored **50+ events** and mentored juniors through technical workshops, outreach charities and club activities.